

The Unified Field Action and Equations of NVG/VMF: Coupling Gravity, Electrodynamics, and the Vacuum Condensate

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Repository: <https://github.com/infosave2007/vmf>

Abstract

We present the complete mathematical formulation and physical verification of the **Unified Field Action and Equations** within the Null-Vector Gravity (NVG) and Vacuum Mass Fraction (VMF) framework. By defining a single action $S[g_{\mu\nu}, \mathcal{W}, \theta, A_\mu]$ that couples General Relativity, Electrodynamics, and the macroscopic vacuum condensate (parameterized by amplitude $\mathcal{W}$ and phase θ), we derive the field equations governing their joint interaction. We show that the cosmological bounce (singularity resolution) and the rotation-independent generation of magnetar magnetic fields (chiral-like instability) emerge naturally as two limiting regimes of these unified equations, requiring no free parameters other than the lattice QCD vacuum anchor $M_{\Omega,0} = 859 \pm 8$ MeV.

1. Introduction and Motivation

In the NVG/VMF program, dense nuclear matter is not treated as moving through a passive, static vacuum. Instead, the non-perturbative QCD contribution to hadronic mass is represented by a macroscopic vacuum order parameter whose in-medium evolution is coupled to both the geometry and the electromagnetic sector.

While previous preprints analyzed the cosmological bounce (Pillar II) and magnetar magnetic fields (Pillar I) as separate systems, a unified field theory is required to prove that they are consistent manifestations of the same underlying action. This document provides the formal derivation and numerical verification of this unified action, demonstrating that a single set of field equations governs the physics of the Big Bang turnaround, black hole singularity resolution, and compact object field generation.

2. The Unified Lagrangian Action

We define the total action of the unified fields as:

$$S[g_{\mu\nu}, \Phi, A_\mu] = \int d^4x \sqrt{-g} [\mathcal{L}_{\text{Grav}} + \mathcal{L}_{\text{Vac}} + \mathcal{L}_{\text{EM}} + \mathcal{L}_{\text{Topo}} + \mathcal{L}_{\text{matter}}]$$

where the components are defined as follows:

1. **Gravitational Lagrangian ($\mathcal{L}_{\text{Grav}}$):** In the NVG/VMF framework, the preferred time direction is defined by the gradient of the Goldstone phase θ of the vacuum condensate. Thus, the unit timelike vector field u_μ is a composite field defined directly as:

$$u_\mu \equiv \frac{\partial_\mu \theta}{\sqrt{-g^{\alpha\beta} \partial_\alpha \theta \partial_\beta \theta}}$$

This composite vector field automatically satisfies the constraint $g^{\mu\nu} u_\mu u_\nu = -1$, removing the need for a dynamical null-vector field or a Lagrange multiplier. The gravitational Lagrangian then reduces to the standard Einstein-Hilbert term:

$$\mathcal{L}_{\text{Grav}} = \frac{R}{16\pi G}$$

2. **Vacuum/Condensate Lagrangian ($\mathcal{L}_{\text{Vac}}$):** The macroscopic vacuum condensate is modeled as a complex scalar field $\Phi(x) = \mathcal{W}(x)e^{i\theta(x)}$, where $\mathcal{W}$ is the condensate amplitude and θ is the Goldstone phase (which acts as the emergent time variable):

$$\mathcal{L}_{\text{Vac}} = -g^{\mu\nu} \partial_\mu \Phi^* \partial_\nu \Phi - V(|\Phi|)$$

Evaluating this in terms of the amplitude and phase yields the standard kinetic components:

$$\mathcal{L}_{\text{Vac}} = -g^{\mu\nu} \partial_\mu \mathcal{W} \partial_\nu \mathcal{W} - \mathcal{W}^2 g^{\mu\nu} \partial_\mu \theta \partial_\nu \theta - V(\mathcal{W})$$

where the Ginzburg-Landau potential drives vacuum symmetry breaking:

$$V(\mathcal{W}) = \frac{\lambda_v}{4} (\mathcal{W}^2 - M_{\Omega,0}^2)^2$$

3. **Electromagnetic Lagrangian ($\mathcal{L}_{\text{EM}}$):** Kalb-Ramond-like coupling where the gauge-kinetic function $Z_{\text{EM}}(\mathcal{W})$ depends on the vacuum condensate density, modifying the effective dielectric constant in dense cores:

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4} Z_{\text{EM}}(\mathcal{W}) F_{\mu\nu} F^{\mu\nu}$$

where $Z_{\text{EM}}(\mathcal{W}) = e^{-2\alpha_v \mathcal{W}/M_W}$.

4. **Topological Lagrangian ($\mathcal{L}_{\text{Topo}}$):** Standard, gauge-invariant axion-like coupling between the vacuum Goldstone phase θ and the electromagnetic field:

$$\mathcal{L}_{\text{Topo}} = \gamma_{\text{topo}} \frac{\alpha_{\text{EM}}}{8\pi} \theta F_{\mu\nu} \tilde{F}^{\mu\nu}$$

where $\tilde{F}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\alpha\beta} F_{\alpha\beta}$ is the dual electromagnetic field strength tensor, γ_{topo} is the dimensionless topological coupling constant, and α_{EM} is the fine-structure constant.

3. Derivation of the Field Equations

Varying the total action with respect to each field yields the four core equations of the unified framework:

3.1. Modified Einstein Equations (Geometry)

Varying the action with respect to the metric $g^{\mu\nu}$:

$$G_{\mu\nu} = 8\pi G (T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\text{Vac}} + T_{\mu\nu}^{\text{EM}})$$

where the vacuum energy-momentum tensor is:

$$T_{\mu\nu}^{\text{Vac}} = 2\partial_\mu \mathcal{W} \partial_\nu \mathcal{W} + 2\mathcal{W}^2 \partial_\mu \theta \partial_\nu \theta - g_{\mu\nu} (\partial^\alpha \mathcal{W} \partial_\alpha \mathcal{W} + \mathcal{W}^2 \partial^\alpha \theta \partial_\alpha \theta + V(\mathcal{W}))$$

At extreme densities ($\rho \rightarrow \rho_c$), the potential $V(\mathcal{W})$ dominates, violating the Strong Energy Condition ($\varepsilon + 3P < 0$) and halting the collapse.

3.2. Modified Maxwell Equations (Electrodynamics)

Varying the action with respect to the vector potential A_μ :

$$\nabla_\nu (Z_{\text{EM}}(\mathcal{W}) F^{\mu\nu}) = 4\pi (J_{\text{matter}}^\mu + J_{\text{topo}}^\mu)$$

where the topological current is generated by the Goldstone phase flow of the vacuum condensate:

$$J_{\text{topo}}^\mu = \gamma_{\text{topo}} \frac{\alpha_{\text{EM}}}{2\pi} \tilde{F}^{\mu\nu} \partial_\nu \theta = \gamma_{\text{topo}} \frac{\alpha_{\text{EM}}}{4\pi} \varepsilon^{\mu\nu\alpha\beta} \partial_\nu \theta F_{\alpha\beta}$$

3.3. Vacuum Gap Equation (Condensate Amplitude)

Varying the action with respect to the amplitude $\mathcal{W}$:

$$\square \mathcal{W} - \mathcal{W} g^{\mu\nu} \partial_\mu \theta \partial_\nu \theta - \frac{1}{2} \frac{\partial V}{\partial \mathcal{W}} - \frac{1}{8} \frac{\partial Z_{\text{EM}}}{\partial \mathcal{W}} F_{\alpha\beta} F^{\alpha\beta} = 0$$

The final term is the **magnetic backpressure** acting directly on the vacuum state. Intense magnetic fields ($B \sim 10^{16}$ G) feed back into the gap equation, shifting the condensate value and stiffening the EoS.

3.4. Topological Phase Conservation

Varying the action with respect to the Goldstone phase θ :

$$\nabla_\mu (\mathcal{W}^2 g^{\mu\nu} \partial_\nu \theta) = J_{\text{source}}^{\text{topo}}$$

This equation shows that the topological phase flow is conserved outside the cores of quantized vacuum vortices, where phase singularities ($\nabla \times \nabla \theta \neq 0$) act as topological source charges.

4. Limits and Astrophysical Verification

The unified equations are verified across three physical regimes via `verification/nvg_unified_field_equations.py`:

4.1. Cosmological Bounce Limit

In the early universe, as density approaches the QCD scale $\rho \approx \rho_c$, the vacuum condensate melts ($\mathcal{W} \rightarrow 0$), and the Ginzburg-Landau potential saturates at $V(0) = \frac{\lambda_v}{4} M_{\Omega,0}^4$. - **Bounce Density:**

$$\rho_c = \frac{M_{\Omega,0}^4}{(\hbar c)^3} \approx 7.09 \times 10^4 \text{ MeV/fm}^3$$

- **Singularity Resolution:** Under FLRW minisuperspace reduction, the backreaction of the melting vacuum field modifies the Friedmann equation to $H^2 = \frac{8\pi G}{3} \rho(1 - \rho/\rho_c)$. At $\rho = \rho_c$, the expansion halts ($H = 0$) and accelerates outwards with a positive turnaround acceleration:

$$\frac{\ddot{a}}{a} = +\frac{16\pi G}{3} \rho_c \approx 1.41 \times 10^{11} \text{ s}^{-2}$$

confirming a smooth, singularity-free semi-classical bounce.

4.2. Magnetar Chiral Instability Limit

During the rapid core collapse phase of a neutron star (duration $t_{\text{trans}} \sim 1 \text{ ms}$), the melting of the vacuum field ($\Delta\mathcal{W} \approx 20\%$) drives a Goldstone phase velocity:

$$\dot{\theta} = \frac{\Delta\mathcal{W}}{\hbar} \approx 2.61 \times 10^{23} \text{ rad/s}$$

Under time-dependent phase approximation, this reduces to the standard chiral magnetic current $\mathbf{J}_{\text{topo}} = \sigma_{\text{topo}} \mathbf{B}$. - **Chiral Magnetic Instability:** In a core with conductivity $\sigma_{\text{core}} \sim 10^{29} \text{ s}^{-1}$, this current triggers a chiral growth rate $\Gamma_{\text{growth}} = \sigma_{\text{topo}}^2 / (4\sigma_{\text{core}})$. - **Amplification Scale:** To achieve the catalog-inferred structural field amplification $\Gamma_{\text{struct}} \approx 5.3$, the required topological coupling constant is:

$$\gamma_{\text{topo}} \approx 7.4 \times 10^{-5}$$

which corresponds to $\sim 6.4\%$ of the standard EM-axion scale $\alpha_{\text{EM}}/(2\pi) \approx 1.16 \times 10^{-3}$, proving that a small, few-percent topological coupling is sufficient to generate magnetar-strength fields without millisecond birth spin.

4.3. Magnetic Backpressure Limit

In the core of a massive post-merger remnant ($B_{\text{core}} \approx 2.82 \times 10^{16} \text{ G}$), the magnetic pressure reaches $P_{\text{mag}} \approx 0.0197 \text{ MeV/fm}^3$. - **Condensate Shift:** The

magnetic backpressure term in the gap equation shifts the vacuum condensate by:

$$\frac{\delta\mathcal{W}}{\mathcal{W}_0} \approx -7.02 \times 10^{-5}$$

This shift is small enough to guarantee standard vacuum stability while providing the necessary EoS stiffening. - **Observational Signature:** This backpressure changes the EoS stiffness, shifting the post-merger gravitational wave f_2 peak frequency by:

$$\Delta f_2 \approx -0.395 \text{ Hz}$$

which is detectable by the Einstein Telescope.

5. Data Availability

The numerical implementations and scripts verifying these equations are contained in the VMF repository: - **Unified Equations Script:** verification/nvg_unified_field_equations.py - **Vortex-Coupling Script:** verification/nvg_magnetar_vortex_coupling.py - **Decoherence and Maxwell Script:** verification/nvg_em_maxwell_decoherence.py - **Predictions Script:** verification/nvg_new_predictions.py - **Master Suite:** verification/run_all_checks.py